

Course Code: CSE 3080	Course Title: Quantum Computing Type of Course: Integrated	L- T- P- C	2 -0	2	3
Version No.	1				
Course Pre-requisites	Linear Algebra Probability and Statistics				
Anti-requisites					
Course Description	This course provides an introduction to the theory and practice of quantum computation. Topics covered include: quantum mechanics to understand quantum computation. Quantum algorithms. The Shor's factorization algorithm Grover's search algorithm Mathematical models of quantum computation, Quantum Machine Learning, and to physical systems.				
Course Objective	The objective of the course is to familiarize the learners with the concepts of Quantum Computing and attain EMPLOYABILITY SKILLS through EXPERIENTIAL LEARNING techniques				
Course Out Comes	On successful completion of the course the students shall be able to: Understand the basic principles of quantum computation and quantum mechanics. Design quantum circuits using quantum gates. Analyze the behavior of basic quantum algorithms. Understand the difference between classical and quantum machine learning approach.				
Course Content:					
Module 1	INTRODUCTION	Quiz	Quiz	10 sessions (8 T + 2 L)	
Topics: Introduction to quantum computing. Qubits, Bloch sphere, multiple qubits, quantum states and measurements, Postulates of quantum mechanics, Classical computation vs quantum computation.					
Module 2	QUANTUM MODEL OF COMPUTATION	Quiz	Quiz	12 sessions (8 T + 4 L)	
Topics: The model of quantum computation, Quantum circuits: single qubit gates, multiple qubit gates, design of quantum circuits.					
Module 3	QUANTUM ALGORITHMS	Assignment	Case Studies	12 sessions (8 T + 4 L)	
Topics: Deutsch-Jozsa algorithm and Grover's search algorithm. Shor's algorithm for factoring, Quantum Fourier transform.					

Module 4	QUANTUM INFORMATION THEORY & QUANTUM MACHINE LEARNING	Assignment	Case Studies	11 sessions (9 T + 2 L)
Topics: Comparison between classical and quantum information theory, Applications of quantum information, Bell states, Quantum Machine Learning, no cloning theorem.				
List of Laboratory Tasks: Lab 1: Use Qiskit Tools [Module 1] Lab 2: Display and Use System Information [Module 1] Lab 3: Construct Visualizations [Module 1] Lab 4: Perform Operations on Quantum Circuits [Module 2] Lab 5: Implement BasicAer: Python-based Simulators [Module 2] Lab 6: Access Aer Provider [Module 3] Lab 7: Implement QASM [Module 3] Lab 8: Executing Experiments [Module 3] Lab 9: Return the Experiment Results [Module 4] Lab 10: Compare and Contrast Quantum Information [Module 4]				
Targeted Application & Tools that can be used Framework- Qiskit Language- Python Applications: Quantum Circuits Quantum Gates Quantum Machine Learning Algorithms				
Project work/Assignment:				
Assignment: Create quantum circuit functions that can compute the XOR, AND, NAND and OR gates using the NOT gate (expressed as x in Qiskit), the CNOT gate (expressed as cx in Qiskit) and the Toffoli gate (expressed as ccx in Qiskit) . Measure the Bloch sphere coordinates of a qubit using the Aer simulator and plot the vector on the Bloch sphere Investigate the relationship between the number of qubits required for the desired accuracy of the phase estimation with high probability. Project Work: Create a program that builds an oracle for a given string (e.g. given 01101, will return a QuantumCircuit that inverts the phase of the state 01101>) and leaves all other states unchanged. Tackle an open issue in the Qiskit Terra repo.				

Create a program that builds an oracle circuit from a problem (like the PhaseOracle class does in the previous page). Assess how the size of your circuits grow with the size of the problem.

Text Book

Nielsen, M., & Chuang, I. (2010). Quantum Computation and Quantum Information: 10th Anniversary Edition. Cambridge: Cambridge University Press. doi:10.1017/CBO9780511976667

McMahon D. Quantum Computing Explained. Hoboken N.J: Wiley-Interscience : IEEE Computer Society; 2008.

References

Benenti G., Casati G. and Strini G., Principles of Quantum Computation and Information, Vol. I: Basic Concepts, Vol II: Basic Tools and Special Topics, World Scientific. (2004)

Pittenger A. O., An Introduction to Quantum Computing Algorithms (2000).

E book link R1:

<http://community.qiskit.org/textbook>

E book link R2

<https://github.com/Qiskit>

R3 Web resources:

Abraham Asfaw and Antonio Corcoles & et al. "Learn Quantum Computation Using Qiskit", 2020, <http://community.qiskit.org/textbook>

IBM Qiskit Global Summer School 2021: Quantum Machine Learning, <https://qiskit.org/events/summer-school/>

<https://quantum-computing.ibm.com/>

<https://qiskit.org/>

<https://presiuniv.knimbus.com/u>

Topics relevant to development of "Employability Skills"

Designing Quantum circuits

Visualizing Quantum Circuit outputs

Analyzing and Comparing Quantum Algorithm Performance for developing Employability Skills through Experiential Learning techniques. This is attained through assessment component mentioned in course handout.